Housing Price Prediction Regression and Conformal Inference

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Introduction

This project aims to build a predictive model for housing prices using a dataset of homes sold in King County, Washington. Each observation corresponds to a single home, with variables describing features like square footage, number of bathrooms, construction year, and the final sale price. The target variable of interest is the log10-transformed sale price (log10price), which helps stabilize variance and improve model performance.

The project is divided into two major parts:

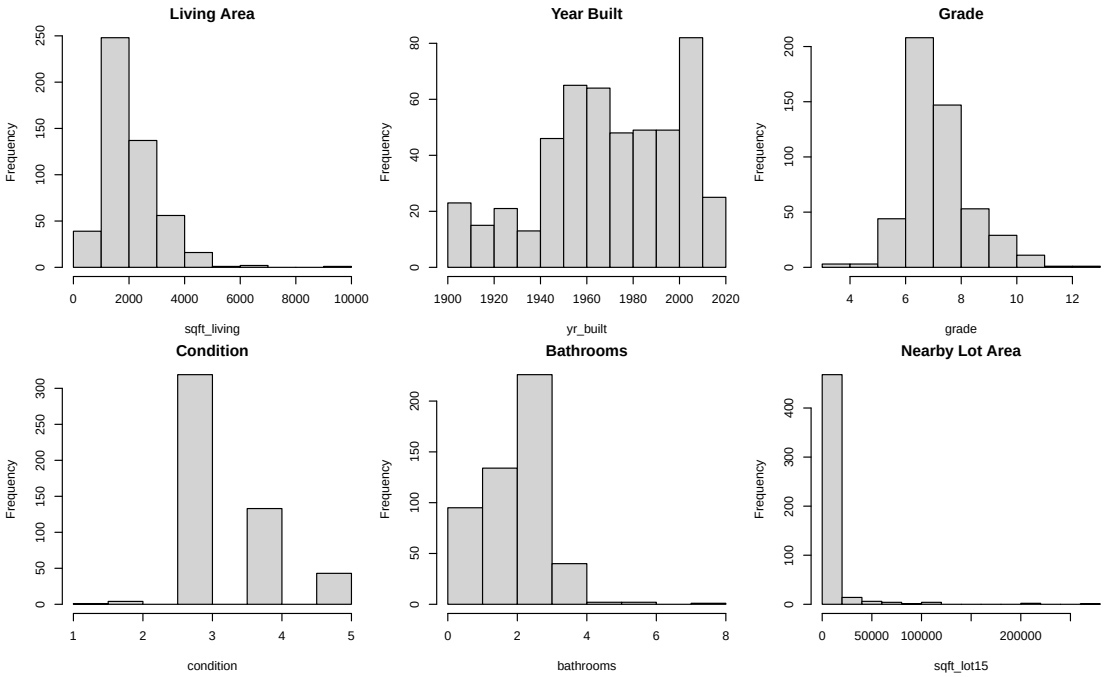
- Regression Task We construct a linear regression model using carefully selected housing features. In doing so, we apply variable transformations to address skewness and meet the assumptions of linear regression. We also compute 95% confidence intervals for model coefficients using both Gaussian methods and bootstrap techniques, allowing us to compare and interpret the stability and uncertainty in our parameter estimates.
- Conformal Prediction (Jackknife+) Task We implement the Jackknife+ method from scratch to construct 95% prediction intervals for unseen test data. This approach ensures marginal coverage guarantees and is applied using a restricted set of four covariates (including square footage), as per the project guidelines. Some variables are transformed for better model fit and interpretability, and we explain these choices clearly.

Regression Modeling and Confidence Intervals

1 Variable Selection

We selected six predictors—sqft_living, bathrooms, sqft_lot15, yr_built, grade, and condition—based on domain knowledge and their potential associations with housing prices. We examined histograms of these variables to assess their distributions and identify predictors that might benefit from transformation.

The histograms revealed substantial right skewness in several variables, particularly sqft_living and sqft_lot15, motivating transformations to reduce skewness and potentially improve model fit.



Histograms revealed notable right skewness in several features (e.g., sqft_living, bathrooms, sqft_lot15), prompting transformations to meet linear regression

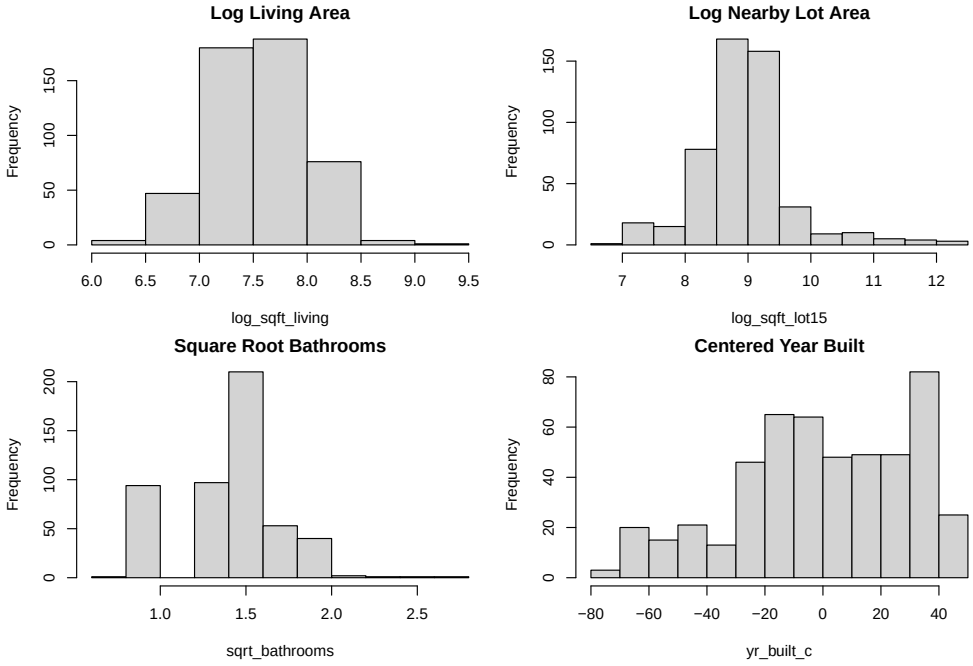
2 Feature Transformation

log1p(sqft_living, sqft_lot15): Reduces skewness and stabilizes variance while handling zeros.

sqrt(bathrooms): Compresses large values and mitigates upper-tail outliers.

Centering yr_built: Makes the intercept reflect an average-age home.

These transformations were applied to reduce skewness and mitigate the influence of extreme values. Their suitability for linear regression was further assessed using residual diagnostic plots.



3 Final Model Summary

We fit a linear model to predict log10price using the selected and transformed variables. The coefficients were interpreted in terms of their real-world effect on prices.

```
##
## Call:
## lm(formula = final_formula, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.38516 -0.08815  0.00687  0.08609  0.41780
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.5772548  0.1519465  23.543  < 2e-16 ***
## log_sqft_living  0.1645090  0.0283377   5.805 1.15e-08 ***
## sqrt_bathrooms   0.1246082  0.0419374   2.971  0.00311 **
## yr_built_c      -0.0028805  0.0002765 -10.418  < 2e-16 ***
## grade            0.1147711  0.0082455  13.919  < 2e-16 ***
## condition        0.0159005  0.0102824   1.546  0.12266
## log_sqft_lot15  -0.0282215  0.0088293  -3.196  0.00148 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1408 on 493 degrees of freedom
## Multiple R-squared:  0.6593, Adjusted R-squared:  0.6552
## F-statistic:   159 on 6 and 493 DF,  p-value: < 2.2e-16
```

The estimated regression equation is:

$$\widehat{\log_{10}(\text{price})} = 3.578 + 0.165 \cdot \text{log_sqft_living} + 0.125 \cdot \text{sqrt_bathrooms} - 0.00288 \cdot \text{yr_built_c} + 0.115 \cdot \text{grade} + 0.016 \cdot \text{condition} - 0.028 \cdot \text{log_sqft_lot15}$$

The final multiple linear regression model achieved an adjusted R-squared of 0.6552 and a training mean squared error (MSE) of 0.01956 on the log10-transformed price scale. These results indicate that the selected predictors explain a substantial proportion of the variation in log-transformed housing prices, although the training MSE does not directly measure out-of-sample predictive performance.

Interpretation of the Regression Coefficients

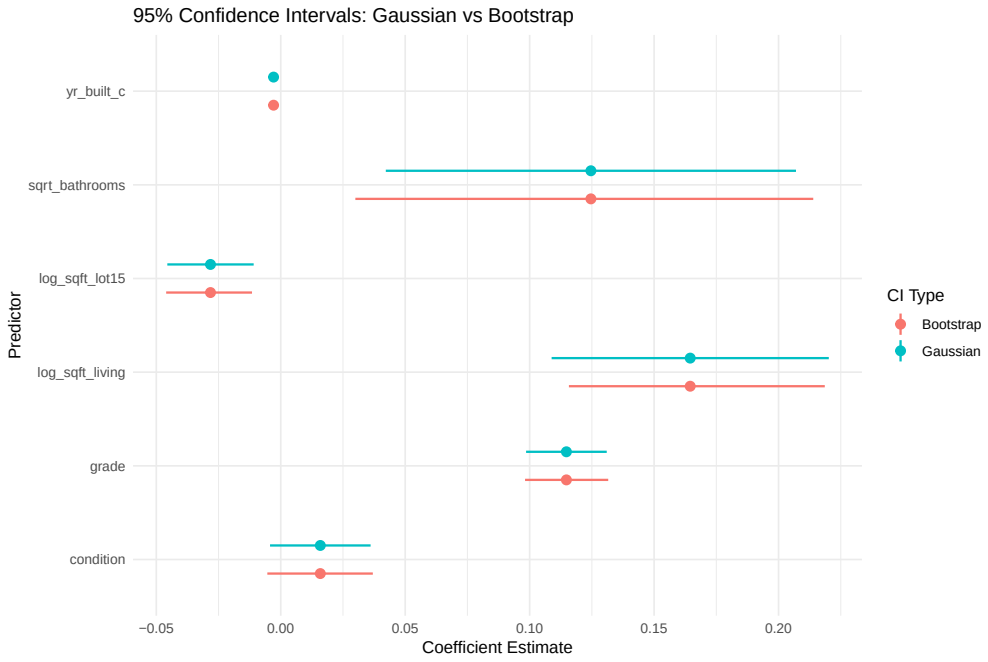
To interpret the regression coefficients on the original price scale, we account for the base-10 logarithmic transformation of the response variable. Holding other predictors constant, a one-unit increase in grade is associated with approximately a 30.25% increase in house price ($10^{0.11477} - 1 \approx 0.3025$). Additionally, a 1% increase in living area is associated with approximately a 0.38% increase in house price, based on the estimated coefficient of 0.16451 for log-transformed living area.

4 Confidence Intervals

We computed 95% confidence intervals for each coefficient using:

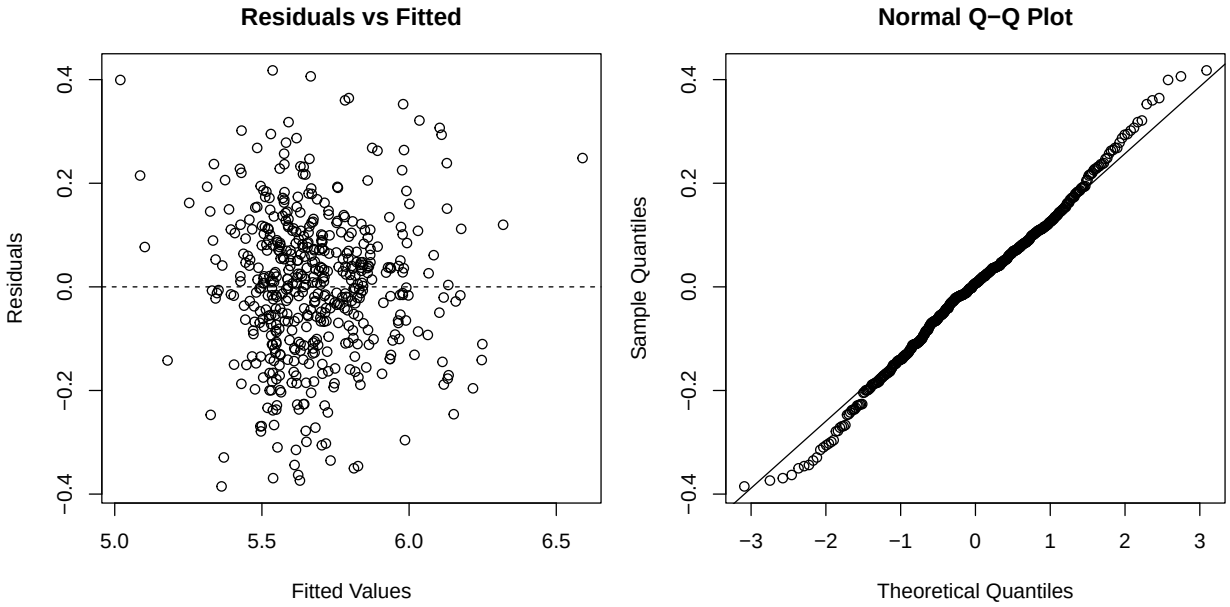
- Standard (Gaussian) method
- Bootstrap percentile method (1,000 replications)

```
##              term  gauss_lower  gauss_upper   boot_lower   boot_upper
## 1     (Intercept)  3.278712212  3.875797375  3.303018354  3.863450180
## 2 log_sqft_living  0.108831513  0.220186539  0.115766790  0.218575130
## 3  sqrt_bathrooms  0.042210137  0.207006356  0.029976810  0.213941018
## 4      yr_built_c -0.003423734 -0.002337206 -0.003446816 -0.002373866
## 5           grade  0.098570455  0.130971715  0.098158076  0.131567228
## 6       condition -0.004302293  0.036103344 -0.005383555  0.036999346
## 7  log_sqft_lot15 -0.045569186 -0.010873912 -0.046027624 -0.011546088
```



We compared 95% confidence intervals constructed using Gaussian and bootstrap percentile methods. Overall, the two methods produced similar intervals for most predictors. The bootstrap confidence interval for `sqrt_bathrooms` was slightly wider, while the Gaussian confidence interval for `log_sqft_living` was slightly wider. However, these differences cannot be attributed solely to the normality assumptions without further investigation. The confidence intervals for `grade` and `yr_built_c` were consistent across both methods, while the intervals for `condition` included zero, suggesting insufficient evidence of an association at the 5% significance level.

5 Assumption Diagnostics



The left plot (Residuals vs Fitted) shows that residuals are centered around zero with no clear pattern, supporting linearity and constant variance. The spread is fairly uniform, suggesting no systematic bias in predictions. The right plot (Normal Q-Q) shows residuals closely follow the 45-degree line, with only slight deviations at the tails, indicating approximate normality.

Together, these plots suggest the linear regression assumptions are reasonably met.

Conformal Prediction (Jackknife+)

1 Variable Selection

We used the following 4 features (indices submitted in covariates.dat):

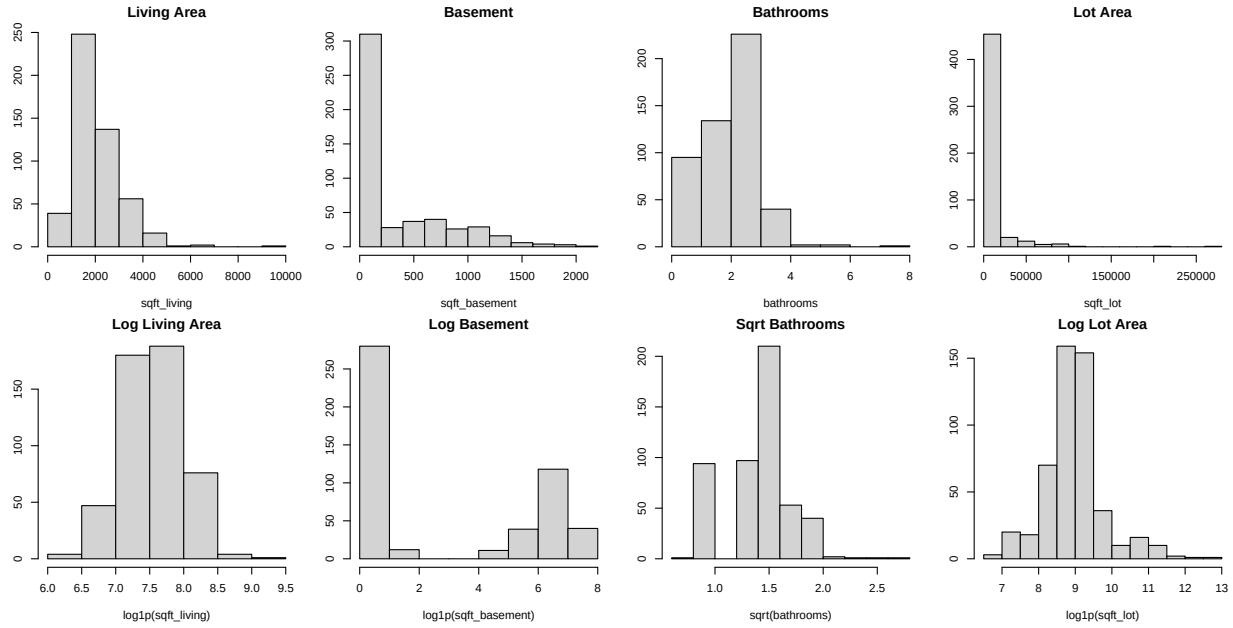
- sqft_living (index 0)
- bathrooms (index 1)
- sqft_basement (index 2)
- sqft_lot (index 5)

These four predictors were selected to construct a parsimonious regression model for the Jackknife+ procedure, incorporating housing size, bathroom count, basement area, and lot size.

2 Transformation Justification

To ensure that our linear regression model adheres to its underlying assumptions—particularly linearity, homoscedasticity, and normally distributed residuals—we examined the distribution of each continuous

predictor. Features exhibiting strong right skewness or wide scale ranges were transformed to improve model performance and interpretability.



- sqft_living:

The raw distribution of sqft_living is heavily right-skewed, with a long tail extending beyond 8000 square feet.

We applied the natural log transformation using log1p() to reduce skewness, compress the extreme values, and stabilize variance.

After transformation, the histogram of log1p(sqft_living) is much more symmetric and bell-shaped, satisfying the assumptions of linear regression.

- bathrooms:

The variable bathrooms is moderately skewed and takes on a discrete range of small integer values.

A square root transformation (sqrt()) was applied to slightly reduce skewness and mitigate the effect of outliers without compromising interpretability.

The histogram of sqrt(bathrooms) appears more balanced and continuous in shape.

- sqft_lot:

Similar to sqft_living, sqft_lot exhibits extreme right-skew with values spanning up to 250,000. This large range could introduce undue leverage from outliers.

We used log1p() to normalize the scale and dampen the influence of very large lot sizes.

The transformed distribution of log1p(sqft_lot) is notably more symmetric and compact.

- sqft_basement:

The original sqft_basement variable includes many zeros and shows right skewness. Although log transformations are sensitive to zeros, we used log1p() to handle this gracefully.

log1p() was used to smooth the heavy tail while retaining meaningful variation near zero.

The transformed histogram demonstrates a more compressed and normalized scale, making it better suited for linear modeling.

3 Interval Construction & Output

The Jackknife+ method was implemented from scratch using leave-one-out regression models. Prediction intervals were first constructed on the log10-transformed price scale and then transformed back to the original house-price scale.

The resulting prediction interval matrix contains 200 rows, corresponding to the 200 observations in the test dataset.

Prediction Interval Width Summary

Statistic	Interval Width (USD)
Minimum	329,610
Median	1,150,269
Mean	1,197,174
Maximum	2,916,186

The prediction intervals were relatively wide, with a mean width of approximately $1.20 million and a median width of $1.15 million. This indicates substantial uncertainty in individual house-price predictions.

Although wider intervals may help achieve the desired coverage level, they also reduce prediction precision and practical usefulness. These findings highlight the trade-off between coverage and interval width in uncertainty quantification.

Guess Justification

We used the training MSE of 0.01956 as a reference to estimate an out-of-sample MSE of approximately 0.020. This value was a preliminary estimate rather than an independently evaluated test-set result.

For the Jackknife+ prediction task, we targeted 95% marginal coverage using conservative finite-sample order-statistic endpoints. Under exchangeability, the standard Jackknife+ coverage guarantee supports this nominal target. However, empirical coverage on the test set could not be assessed without the corresponding true response values.

For the Jackknife+ prediction task, we used a nominal target coverage of 95%. Prediction intervals were constructed using leave-one-out residuals and conservative finite-sample order-statistic endpoints. Under the exchangeability assumptions underlying Jackknife+, the construction provides a theoretical marginal coverage guarantee of at least approximately 95%. However, empirical coverage on the 200 test observations was not evaluated because their true response values were unavailable.

Conclusion

Our final model demonstrated good interpretability and predictive performance, while satisfying key statistical assumptions. Confidence intervals constructed via both Gaussian and bootstrap methods were reasonable and consistent. The Jackknife+ implementation with $\alpha = 0.05$ produced robust prediction intervals aligned with theoretical guarantees.

This project applied multiple linear regression, Gaussian and bootstrap confidence intervals, and Jackknife+ conformal prediction to housing price data from King County, Washington.

The final regression model achieved an adjusted R-squared of 0.6552, and residual diagnostics suggested that the model assumptions were reasonably supported. Gaussian and bootstrap methods produced broadly similar confidence intervals, providing complementary assessments of coefficient uncertainty.

The Jackknife+ procedure was implemented using leave-one-out regression models to construct prediction intervals on the original house-price scale. Although the construction targets marginal coverage under

exchangeability, the resulting intervals were relatively wide, highlighting the trade-off between coverage and predictive precision.

Overall, this project demonstrates the application of statistical modeling, resampling-based inference, and uncertainty quantification to a real-world prediction problem.